DFA DESIGN

LOGIN BRUTE-FORCE PROTECTOR

Theory of DFA

A Deterministic Finite Automaton (DFA) is a mathematical model used to recognize patterns in input strings. It consists of a finite set of states, an input alphabet, a transition function, an initial state, and a set of final states.

A DFA is formally represented as:

$M = (Q, \Sigma, \delta, q_0, F)$

Where:

- Q = Set of states.
- Σ = Input alphabet.
- δ = Transition function.
- q_0 = Initial state.
- F = Set of final states.

In this project, the DFA processes each login attempt as an input symbol and changes its state according to whether the credentials are correct or incorrect.

States of the DFA

The DFA consists of five states:

- START (S0): The initial state before any failed login attempt.
- ONE_FAIL (S1): Represents one consecutive failed login attempt.
- TWO_FAIL (S2): Represents two consecutive failed login attempts.
- LOCKED (S3): Represents three consecutive failed attempts. Further login attempts are rejected.
- SUCCESS (S4): Represents a successful login.

Input Alphabet

The input alphabet contains two symbols:

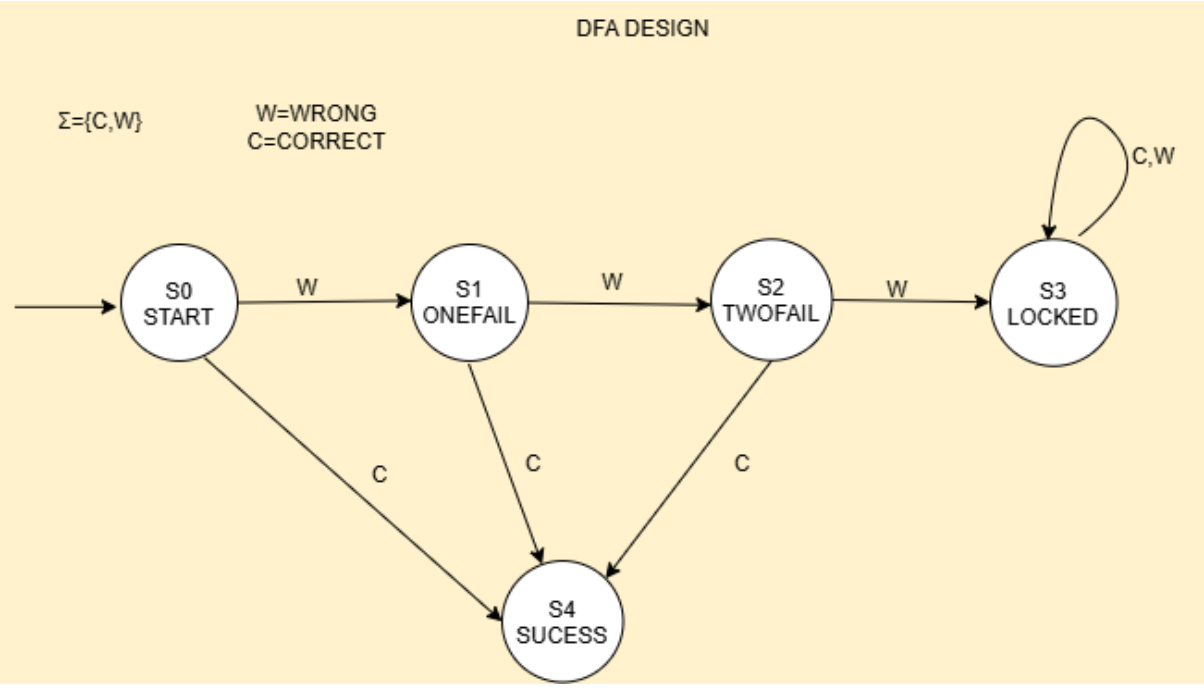
$\Sigma = \{C, W\}$

Where:

- C = Correct username and password.
- W = Wrong username or password.

- The DFA reads one input symbol for every login attempt and moves to the corresponding next state.

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Transition Function

The transition function determines the next state based on the current state and the input symbol.

The transition table is given below.

Current State	Correct (C)	Wrong (W)
START (S0)	SUCCESS (S4)	ONE_FAIL (S1)
ONE_FAIL (S1)	SUCCESS (S4)	TWO_FAIL (S2)
TWO_FAIL (S2)	SUCCESS (S4)	LOCKED (S3)
SUCCESS (S4)	SUCCESS (S4)	ONE_FAIL (S1)
LOCKED (S3)	LOCKED (S3)	LOCKED (S3)